

THE PRIME-WAVE ANATOMY OF THE GAP-AMPLITUDE LAW FOR RIEMANN ZEROS

UĞUR SEZEN

ABSTRACT. In a companion note we measured the joint law of consecutive-zero gaps δ_n and between-zero maxima $M_n = \max |Z(t)|$ of Hardy’s function, finding that its correlation matches CUE characteristic polynomials only at a shifted effective dimension, with different shifts for different statistics. Here we dissect the deviation. Regressing $\log M_n$ and $\log \delta_n$ on the explicit-formula waves $\cos(t \log p^k)$ (phases evaluated at gap midpoints, exogenous to the amplitude measurement; placebo controls at the noise floor), we find: (i) a *sum rule* — the total prime content of the amplitudes tracks the explicit-formula weights $p^{-k/2}/k$ term by term across a factor 20 in weight, at an overall normalization of 0.95–1.1 whose exact value is frame-dependent at the 5–15% level; (ii) the content splits between a direct amplitude channel w and a zero-displacement channel v , mirror curves that each collapse onto one-variable laws in $\tau = \log p / \log \frac{t}{2\pi}$; (iii) stripping the measured waves from both series leaves a core pair with correlation $r^* \approx 0.97$ – 0.98 — tighter than any CUE in the sampled range $N \in [5, 20]$ — with the Pearson/Spearman effective-dimension anomaly of the companion note melting away entirely; (iv) an exhaustion analysis reveals a role-crossover at $\tau^* \approx 0.40$ and yields an upper bound on the intrinsic (non-arithmetic) variance of the amplitude sequence, $V_{\text{res}}(L)$, which grows at a rate close to the CUE rate but at one sixth its magnitude. All of these laws, fitted at $t \leq 1.6 \times 10^6$, are then re-tested out of sample at the 10^{12} -th zero ($t \approx 2.7 \times 10^{11}$) using Odlyzko’s tables: the τ -laws, the sum rule, the crossover, and the V_{res} extension all hold across this 1.6×10^5 -fold jump in height; the zero-displacement law is further verified at the 10^{21} -st and 10^{22} -nd zeros, where only gap data exist, extending its validated range to eighteen orders of magnitude in t . At single-gap scale, two thirds or more of the apparent randomness of the maxima sequence is deterministic prime signal.

1. INTRODUCTION

Let γ_n denote consecutive ordinates of nontrivial zeros of ζ , $\delta_n = \gamma_{n+1} - \gamma_n$, $M_n = \max_{\gamma_n \leq t \leq \gamma_{n+1}} |Z(t)|$, and $L = \log \frac{t}{2\pi}$. In a companion note [7] we measured the correlation of the unfolded pair $(\delta_n, \tilde{M}_n)$ over 3.4×10^5 gaps up to $t = 1.6 \times 10^6$, validated the dataset against the proven mean-value results of Conrey–Ghosh [2] and Hughes–Lugmayer–Pearce–Crump [4], and compared it with the CUE characteristic-polynomial model [5]. The outcome was an effective-dimension description with a metric-dependent shift: the rank-based (Spearman) effective dimension is constant, $N_{\text{eff}} = L + 2.0$, while the Pearson-based one drifts from $L + 0.75$ to $L + 1.30$ over the measured range, the two differing by 0.93 ± 0.10 throughout — so no single finite CUE reproduces the joint law.

This note asks *what, concretely, the deviation is*, and answers with a set of regression measurements against the explicit-formula frequencies. The toolbox is elementary (ordinary least squares on 10^5 -size samples), but three design choices matter: all wave phases are evaluated at gap midpoints, which are determined by the zeros alone and are therefore exogenous to the amplitude measurements; every stripping experiment is accompanied by a

placebo run with an equal number of non-prime-power frequencies; and no closed forms are fitted — all comparisons are against measured or proven quantities.

2. DATA AND CONVENTIONS

The dataset is that of [7]: six windows of 4×10^4 gaps centered at $t = 1.2, 2.0, 3.5, 6.0, 10, 16 \times 10^5$ ($L = 9.86$ to 12.45), computed by a vectorized Riemann–Siegel engine validated to 7.6×10^{-6} pointwise and cross-checked against the proven mean of M_n^2 (slope within 0.05%, constant term within 1%). Unfolding uses known scales:

$$\tilde{\delta} = \delta \cdot \frac{L}{2\pi}, \quad \tilde{M} = \frac{M}{\sqrt{AL + \alpha_{-1} + 2A + (\alpha_0 + \alpha_{-1})/L}},$$

with $A = (e^2 - 5)/2$ and the α_i of [4].

For a set $\mathcal{Q}$ of prime powers we regress

$$\log \tilde{M}_n \sim 1 + \sum_{q \in \mathcal{Q}} [a_q \cos(t_n \log q) + b_q \sin(t_n \log q)], \quad \log \tilde{\delta}_n \sim (\text{same basis}),$$

with t_n the midpoint of gap n . “Stripping” means subtracting the fitted wave part from both series and re-exponentiating. Placebo runs replace $\mathcal{Q}$ by an equal number of frequencies bounded away from all $\log p^k$; for the ≤ 23 -frequency bases below the placebo changes the correlations by $\sim 10^{-4}$; for the larger exhaustion bases the floor grows to the 10^{-3} level and results are quoted against it.

3. THE SUM RULE

The explicit formula assigns the wave $\cos(t \log p^k)$ weight $p^{-k/2}/k$ in $\log |\zeta(\frac{1}{2} + it)|$. Define the normalized content $u_q = \sqrt{a_q^2 + b_q^2} / (p^{-k/2}/k)$ for $q = p^k$.

Observation 1 (Sum rule). *For all 23 frequencies tested — the primes 2 through 47 and the prime powers 4, 8, 9, 16, 25, 27, 32, 49 — the measured total content of the maxima tracks the explicit-formula weight, which itself spans a factor of 20 (0.71 down to 0.035). In the midpoint frame $|u_q| = 1.01$ to 1.12. The sine components vanish (phase-aligned waves), and placebo frequencies give $|u| \approx 0$.*

The overall normalization carries a *frame systematic*: the gap midpoints are themselves displaced by the prime waves (the v channel of the next section), so they carry phase jitter (rms ≈ 0.14 , a quarter of the mean gap). Re-evaluating the phases at jitter-free positions — the Riemann–von Mangoldt flow started at the window’s first zero — moves every prime to $|u_p| = 0.95$ – 0.97 with *no* frequency drift (the apparent +1% to +12% drift of the midpoint frame is a frame artifact), while the low-weight prime powers become noisier but remain consistent with unity. We therefore state the sum rule as: term-by-term proportionality to $p^{-k/2}/k$ is frame-robust; the overall level is ≈ 1 within a 5–15% frame systematic, the jitter-free frame preferring ≈ 0.96 — intriguingly close to the $\tau \rightarrow 0$ transmission limit measured below.

4. TWO CHANNELS AND THE τ -LAWS

The unit total splits into two measurable channels: a direct amplitude channel (coefficient w_p : the response of $\log \tilde{M}$ at fixed observed gap) and a zero-displacement channel (coefficient v_p : the response of $\log \tilde{\delta}$). Measured across the six windows, the two run in opposite directions:

p	2	3	5	7	11	13
w_p	0.76	0.65	0.50	0.41	0.30	0.26
$ v_p $	0.13	0.20	0.29	0.35	0.42	0.45

and both collapse, across all windows and primes simultaneously, onto single curves in the Berry variable $\tau = \log p/L$ [1] (residual RMS ≈ 0.01 for w). Long waves (small τ) transmit into amplitude; short waves are absorbed into zero positions. The channels are roughly consistent with the sum rule through the gap–amplitude coupling: $w_p + \beta v_p$ lies within 5–30% of unity (the discrepancy growing with p), with $\beta \approx 2.3$ the log-log regression slope of amplitude on gap. We read this as a measured, observable-level decomposition of the hybrid Euler–Hadamard product of Gonek–Hughes–Keating [3]: the “zeros part” and “primes part” are here separated channel by channel, with the partition governed by τ alone.

5. STRIPPING: THE ARITHMETIC CORE

Removing the fitted waves from both series (basis: all $p^k \leq Q$) and recomputing the correlation of the residual pair gives, at $Q = 49$, $r_P^* \approx 0.97$, $r_S^* \approx 0.98$ — far tighter than any CUE in the range $N \in [5, 19]$ (max 0.93) — while the placebo leaves the raw values 0.75/0.88 untouched. Two structural consequences follow.

Observation 2 (The effective-dimension anomaly is arithmetic). *After stripping, the Pearson and Spearman statistics of the residual pair coincide (both 0.9734 at $L = 11.47$; raw discrepancy 0.12 shrinks to 0.008). The two-metric anomaly of [7] is entirely a prime-wave effect.*

Observation 3 (Exhaustion curve). *Extending the basis to all $p^k \leq Q$ with Q up to 300 (79 frequencies; placebo floor at the 10^{-3} level for these large bases, against which the effects below remain large), the stripped correlation $r^*(Q)$ rises to a maximum and then declines. The maximum sits at $\log Q_{\text{peak}}/L \approx 0.40 \pm 0.03$ across windows (the Q grid is coarse): below this threshold the waves act as amplitude noise (stripping tightens the pair), above it as common-mode signal shared by gaps and maxima (stripping removes genuine coupling). We denote this role-crossover $\tau^* \approx 0.40$.*

At $Q = 300$ the stripped share of the amplitude log-variance reaches 82–87%, leaving a residual

$$V_{\text{res}}(L) : \quad 0.163 \ (L=9.86) \rightarrow 0.244 \ (L=12.45),$$

increasing approximately linearly with slope ≈ 0.031 per unit L . For comparison, the CUE amplitude log-variance grows at ≈ 0.036 per unit N but is six times larger in magnitude at matched dimension. V_{res} is an *upper bound* on the intrinsic randomness of the maxima sequence: our per-frequency coefficients are gap-independent, whereas the conditional-variance anatomy shows the transmission grows with gap width, so arithmetic residue survives the stripping.

The $\tau \rightarrow 0$ limit. A dedicated small- τ campaign — four additional windows at $t = 10^8, 10^9, 10^{10}, 10^{11}$ (2×10^4 – 3×10^4 gaps each; the 10^{11} window computed with an anchor-expanded engine after a double-precision phase-noise artifact was caught and eliminated by a placebo-style consistency check against the Odlyzko window) — yields a 44-point table of $w(p, L)$ reaching $\tau_2 = 0.028$. Two findings. First, the τ -collapse acquires a resolvable fine structure: at fixed τ , larger primes transmit slightly less (e.g. at $\tau \approx 0.058$, $w_2 = 0.771$ versus $w_3 = 0.753$, a 4σ split); a secondary variable beyond τ exists, plausibly related to the u_q drift noted above. Second, extrapolating $\tau \rightarrow 0$ under five model variants (pure quadratic; quadratic plus $1/p$; plus $1/L$; $p = 2$ points only; extended prime set) gives $w(0)$ between 0.88 and 0.98, with *every* variant below unity, though at varying strength — the cleanest single-prime variant ($p = 2$ only) is compatible with unity at 1.5σ , while the multi-prime variants exclude it at 4.7σ or more:

Observation 4 (Incomplete transparency). *The amplitude-channel transmission does not reach unity as $\tau \rightarrow 0$: a persistent absorption of 2–12% survives even for the longest prime waves. The exact value is currently blocked by a degeneracy between $1/p$ and $1/L$ corrections (at fixed τ these are tied through $\log p = \tau L$).*

6. OUT-OF-SAMPLE VALIDATION AT THE 10^{12} -TH ZERO

All laws above were extracted from heights $t \leq 1.6 \times 10^6$. Using Odlyzko’s table of the 10^4 zeros following the 10^{12} -th [6] ($t \approx 2.677 \times 10^{11}$, $L = 24.475$), we recomputed the entire battery at a height larger by a factor of 1.6×10^5 . Amplitudes were evaluated by an anchor-expanded Riemann–Siegel engine (206,393-term main sum; base phases $\theta(t_0) - t_0 \log n \bmod 2\pi$ computed once in high precision, increments in double precision); at the tabulated zeros the engine gives median $|Z| = 1.4 \times 10^{-9}$ against between-zero maxima of order 3, validating it to nine digits.

The results, with the small-height laws quoted as predictions:

- *τ -law for w :* the linear guide $0.940 - 2.967\tau$, fitted only to small heights, predicts the six measured w_p at $L = 24.475$ to within 0.02–0.07; e.g. $w_2 = 0.871 \pm 0.007$ against the guide value 0.856. The residuals grow mildly with τ (the true curve is not exactly linear), but the collapse variable survives the jump.
- *Sum rule:* $|u_q| = 0.99$ – 1.07 across the twelve frequencies tested (uncertainties ± 0.02 at $q = 2$ growing to ± 0.2 at the largest q ; the $q = 49$ value 1.18 is 0.9σ from unity).
- *Role-crossover:* at this L the crossover law places $Q_{\text{peak}} = e^{0.40L} \approx 2 \times 10^4$, so the entire accessible range $Q \leq 300$ ($\tau \leq 0.233$) should lie *before* the peak: $r^*(Q)$ should rise without declining. It does ($0.901 \rightarrow 0.959$), in contrast to the peak-and-decline seen at $L = 12.45$ over the same Q range — a successful qualitative out-of-sample prediction.
- *Intrinsic variance:* the linear extension of $V_{\text{res}}(L)$ predicts 0.617 at this height; the measured value is 0.650, high by 5% in the direction expected from the (necessarily) incomplete stripping at this L .
- *Effective dimension:* the N_{eff} test is *inconclusive* here, neither confirming nor refuting the constant shifts: $r_{\text{CUE}}(N)$ flattens at large N ($dr/dN \approx 0.008$), so with 10^4 gaps the uncertainty on N_{eff} is ≈ 0.8 . The raw deficit $r_\zeta - r_{\text{CUE}}(N=L) = -0.014 \pm 0.006$ is consistent with the small-height value -0.019 .

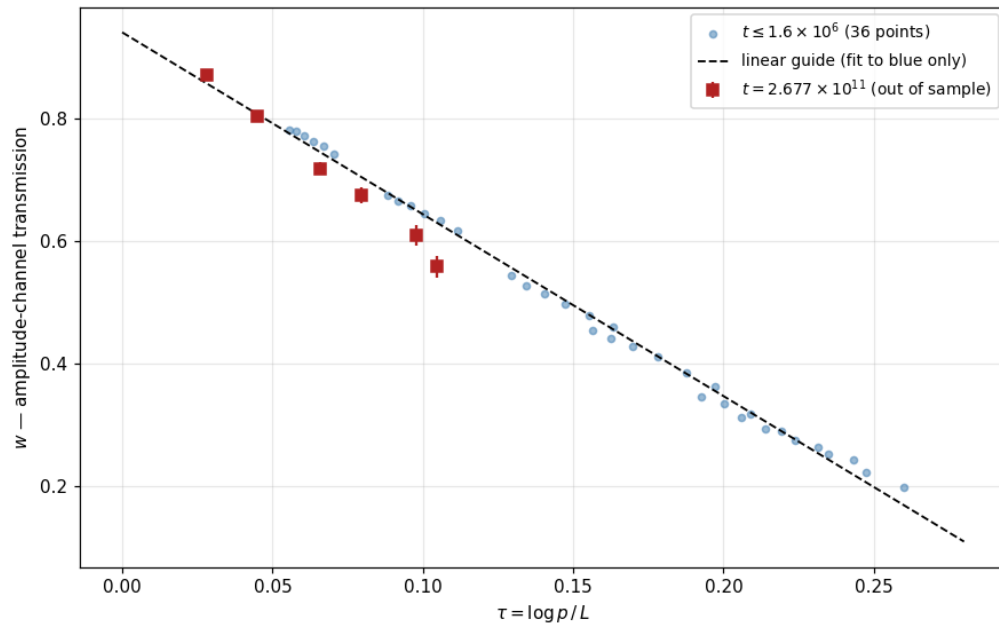


FIGURE 1. The amplitude-channel transmission w against $\tau = \log p/L$. Blue: the 36 measurements at $t \leq 1.6 \times 10^6$. Red: the six out-of-sample measurements at $t = 2.677 \times 10^{11}$. Dashed: the linear guide fitted only to the blue points.

7. THE GAP CHANNEL AT THE 10^{21} -ST AND 10^{22} -ND ZEROS

At heights $t \sim 10^{20}$ the Riemann–Siegel main sum has $\sim 10^{10}$ terms and amplitude computation is out of reach. The zero-displacement channel, however, requires only zero *locations*. Using Odlyzko’s tables of the 10^4 zeros following the 10^{21} -st and 10^{22} -nd zeros [6] ($L = 44.58$ and 46.83 ; stated accuracy 10^{-6} ; wave phases anchored at the tables’ integer base offsets), we regressed the unfolded gap sequence on the six prime waves. All twelve (prime, height) coefficients are detected at $3\text{--}5\sigma$ and agree with the small- τ guide $v = 2.014\tau$ — fitted only to the $t \leq 2.7 \times 10^{11}$ data — within their uncertainties; the most distant point gives $v_2 = 0.0297 \pm 0.0095$ against the predicted 0.0298 . Placebo frequencies sit at the noise floor (Figure 2).

The gap-channel τ -law is thereby verified over *eighteen orders of magnitude* in height: re-measuring the channel on the low-height dataset of [7] extends it down to $t \approx 1.1 \times 10^3$, and the present tables carry it to 1.4×10^{21} . The low windows also reach τ up to 0.46 , where the v -curve is seen to bend away from its linear onset and saturate near $0.55\text{--}0.60$ — in the same region as the role-crossover $\tau^* \approx 0.40$ of the exhaustion analysis, as one would expect if absorption is nearly complete there. We also note the clean linear onset $v \approx 2\tau$ as $\tau \rightarrow 0$, a first coefficient for any future screening theory of the zero gas.

8. WHAT THIS SAYS, AND WHAT IT DOES NOT

At single-gap scale, the “randomness” of $|Z|$ between zeros is, to two thirds or more, deterministic prime signal carried at full explicit-formula weight; the celebrated CUE resemblance of the gap–amplitude law is a single-statistic compensation, not a model equivalence. None of this bears directly on the Riemann Hypothesis, and none of it is a closed-form claim:

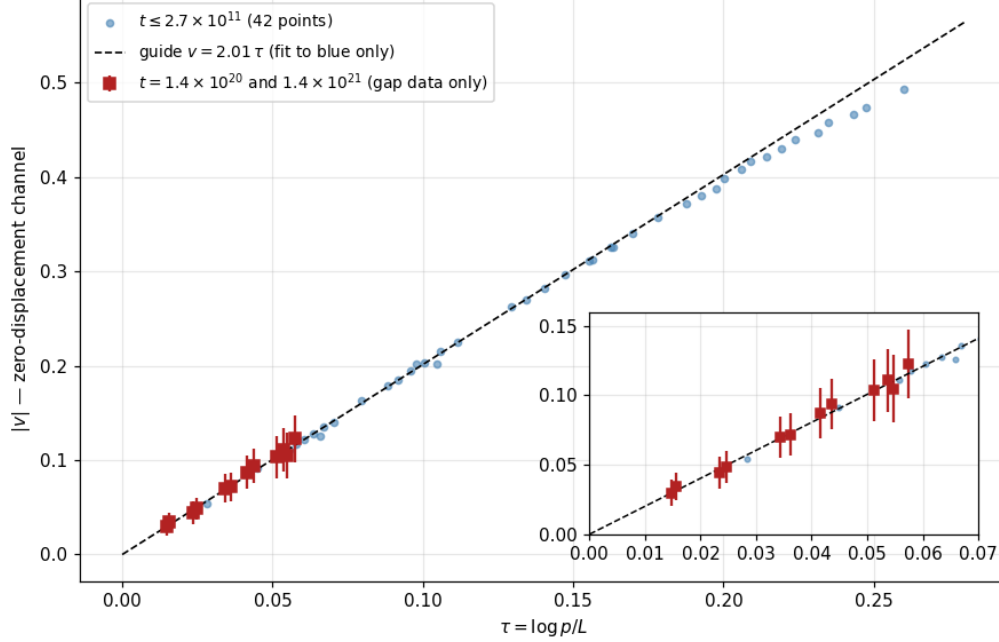


FIGURE 2. The zero-displacement transmission $|v|$ against $\tau = \log p/L$. Blue: measurements at $t \leq 2.7 \times 10^{11}$. Red: the twelve gap-only measurements at $t = 1.4 \times 10^{20}$ and 1.4×10^{21} . Dashed: the linear guide fitted only to the blue points. Inset: the small- τ region.

the two candidate closed forms this project produced at smaller heights were both refuted by larger- T tests and are recorded as such in the project log. What survives are measured curves — $w(\tau)$, $v(\tau)$, the sum rule $u \approx 1$, the crossover $\tau^* \approx 0.40$, and $V_{\text{res}}(L)$ — each stable across a factor 13 in height within the primary windows (and far beyond in the out-of-sample tests), each protected by placebo controls.

Open problems: (1) derive $w(\tau)$ and $v(\tau)$ — equivalently the partition of the unit explicit-formula weight — from a screening theory of the zero gas; (2) explain $\tau^* \approx 0.40$; (3) determine the nature of V_{res} , whose CUE-like growth rate at reduced magnitude suggests a small genuinely-random component of the zero gas; (4) account for the +1% to +12% drift of u_q and the fixed- τ per-prime splitting, plausibly one phenomenon; (5) break the $1/p$ -versus- $1/L$ degeneracy and pin the residual absorption $1 - w(0) \in [0.02, 0.12]$; (6) recover the discriminating power of the N_{eff} test at large heights, which needs $\gtrsim 10^5$ consecutive zeros there (e.g. from the Platt/LMFDB datasets).

REPRODUCIBILITY

All measurements are scripts 27–60 in <https://github.com/azizran/deney> (directory `qm_riemann/`), runnable from the stored gap-maximum datasets; the dated research log in the same directory records the failed intermediate models (independent-phase and common-phase generative attempts) alongside the results reported here.

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REFERENCES

- [1] M. V. Berry, *Semiclassical formula for the number variance of the Riemann zeros*, Nonlinearity **1** (1988), 399–407.
- [2] J. B. Conrey, A. Ghosh, *A mean value theorem for the Riemann zeta-function at its relative extrema on the critical line*, J. London Math. Soc. (2) **32** (1985), 193–202.
- [3] S. M. Gonek, C. P. Hughes, J. P. Keating, *A hybrid Euler–Hadamard product for the Riemann zeta function*, Duke Math. J. **136** (2007), 507–549.
- [4] C. Hughes, S. Lügmer, A. Pearce-Crump, *The second moment of the Riemann zeta function at its local extrema*, arXiv:2411.05573 (2024); J. London Math. Soc. (2) (2025), e70250.
- [5] J. P. Keating, N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Comm. Math. Phys. **214** (2000), 57–89.
- [6] A. M. Odlyzko, *Tables of zeros of the Riemann zeta function*, https://www-users.cse.umn.edu/~odlyzko/zeta_tables/.
- [7] U. Sezen, *The joint law of zero gaps and local maxima of Hardy's Z-function: a numerical study and a CUE effective-dimension anomaly*, draft (2026), same repository.

INDEPENDENT RESEARCHER, ISTANBUL, TURKEY

Email address: ugursezen@gmail.com